

4.1: Matrix Algebra

- [3pts] 12) Let A be an $n \times n$ invertible matrix, and let u and v be two vectors in $\mathbb{R}^n$. Find the necessary and sufficient conditions on u and v in order that the matrix

$$M = \begin{bmatrix} A & u \\ v^T & 0 \end{bmatrix}$$

be invertible, and give a formula for the inverse when it exists.

Solution:

Let A be an $n \times n$ invertible matrix, and let u and v be two vectors in $\mathbb{R}^n$. For M , as defined above, to be an invertible matrix, then $\det M \neq 0$. Further, since A is an invertible matrix, then the following holds:

- A^{-1} exists,
- $\det A \neq 0$,
- $\text{rank}(Au) = n$ for $u \in \mathbb{R}^n$

Next, observe the following Block Gaussian Elimination (manipulation) to make it “simpler” to determine the determinant since we want $\det(M) \neq 0$:

$$\begin{aligned} & \xrightarrow{R_2 - v^T A^{-1} R_1} \begin{bmatrix} A & u \\ v^T - v^T A^{-1}(A) & 0 - v^T A^{-1}u \end{bmatrix} \\ \implies M &= \begin{bmatrix} A & u \\ 0 & -v^T A^{-1}u \end{bmatrix} \end{aligned}$$

Thus, the necessary and sufficient condition for M to be invertible must be for $\text{rank}(M) = n + 1$. That is,

$$\text{rank}(M) = n + 1 \iff v^T A^{-1}u \neq 0$$

To determine the inverse, let us define M^{-1} as follows:

$$M^{-1} = \begin{bmatrix} B & d \\ e^T & c \end{bmatrix}$$

where d, e are vectors in $\mathbb{R}^n$. Then $MM^{-1} = I$ would yield the following:

$$\begin{aligned} MM^{-1} &= I_{n+1} = \begin{bmatrix} A & u \\ 0 & -v^T A^{-1}u \end{bmatrix} \begin{bmatrix} B & d \\ e^T & c \end{bmatrix} \\ \implies \begin{bmatrix} I_n & 0 \\ -v^T A^{-1} & 1 \end{bmatrix} \begin{bmatrix} A & u \\ v^T & 0 \end{bmatrix} \begin{bmatrix} B & d \\ e^T & c \end{bmatrix} &= \begin{bmatrix} I_n & 0 \\ -v^T A^{-1} & 1 \end{bmatrix} \cdot (I_{n+1}) \\ \begin{bmatrix} A & u \\ -v^T + v^T & -v^T A^{-1}u \end{bmatrix} \begin{bmatrix} B & d \\ e^T & c \end{bmatrix} &= \begin{bmatrix} I_n & 0 \\ -v^T A^{-1} & 1 \end{bmatrix} \\ \begin{bmatrix} A & u \\ 0 & -v^T A^{-1}u \end{bmatrix} \begin{bmatrix} B & d \\ e^T c \end{bmatrix} &= \begin{bmatrix} I_n & 0 \\ -v^T A^{-1} & 1 \end{bmatrix} \\ \begin{bmatrix} AB + ue^T & Ad + uc \\ (-v^T A^{-1}u)e^T & (-v^T A^{-1}u)c \end{bmatrix} &= \begin{bmatrix} I_n & 0 \\ -v^T A^{-1} & 1 \end{bmatrix} \end{aligned}$$

We can determine an expression to each entry of M^{-1} in terms of the given M as follows:

$$\begin{aligned} (-v^T A^T u)c &= 1 \implies c = -\frac{1}{v^T A^{-1}u} \\ (-v^T A^{-1}u)e^T &= -v^T A^{-1} \implies e = \frac{A^{-T}v}{v^T A^{-1}u} \\ Ad + uc &= 0 \implies d = \frac{A^{-1}u}{v^T A^{-1}u} \\ AB + ue^T &= I_n \implies B = A^{-1} - \frac{(A^{-1}u)(v^T A^{-1})}{v^T A^{-1}u} \end{aligned}$$

Thus, to put everything back into the matrix M^{-1} :

$$M^{-1} = \begin{bmatrix} A^{-1} - (A^{-1}u)(v^T A^{-1})/v^T A^{-1}u & A^{-1}u/v^T A^{-1}u \\ A^{-T}v/v^T A^{-1}u & -1/v^T A^{-1}u \end{bmatrix}$$

□

- [2pts] 17) A square matrix A is said to be skew-symmetric if $A^T = -A$. Prove that if A is skew-symmetric, then $x^T Ax = 0$ for all x .

Solution:

Let $A \in M_{n \times n}$ (A is a square matrix). Then suppose A is skew-symmetric, then $A^T = -A$. Let x be a vector in $\mathbb{R}^n$, then $x^T Ax$ is of size 1×1 . Using linear algebra manipulations to yield the following:

$$\begin{aligned} x^T Ax &= (x^T Ax)^T \\ &= x^T A^T x \\ &= -x^T Ax \\ 2x^T Ax &= 0 \\ \implies x^T Ax &= 0 \end{aligned}$$

as desired.

Alternatively, the same result could be derived by using the fact that A is skew-symmetric and manipulate the expression algebraically:

$$\begin{aligned} 2x^T Ax &= x^T Ax + x^T Ax \\ &= x^T Ax - x^T A^T x \\ &= x^T Ax - (x^T Ax)^T \\ &= 0 \end{aligned}$$

□

- [3pts] 19) (Continuation) Let A be any square matrix, and define $A_0 = \frac{1}{2}(A + A^T)$ and $A_1 = \frac{1}{2}(A - A^T)$. Prove that A_0 is symmetric, that A_1 is skew-symmetric, that $A = A_0 + A_1$ and that for all x , $x^T Ax = x^T A_0 x$. This explains why, in discussing quadratic forms, we can confine our attention to **symmetric matrices**.

Solution:

Let A be any square matrix, and define $A_0 = \frac{1}{2}(A + A^T)$ and $A_1 = \frac{1}{2}(A - A^T)$. There are **three** parts to prove:

- (a) Prove A_0 is symmetric
- (b) Prove A_1 is skew-symmetric
- (c) Prove $A = A_0 + A_1$ and that $x^T Ax = x^T A_0 x$ for all x .

To prove A_0 is symmetric, it suffices to show that $A_0^T = A_0$:

$$\begin{aligned} A_0 &= \frac{1}{2}(A + A^T) \\ A_0^T &= \left[\frac{1}{2}(A + A^T) \right]^T \\ &= \frac{1}{2}(A + A^T)^T \\ &= \frac{1}{2}(A^T + A) \\ &= A_0 \end{aligned}$$

To prove A_1 is skew-symmetric, it suffices to show that $A_1^T = -A_1$:

$$\begin{aligned} A_1 &= \frac{1}{2}(A - A^T) \\ A_1^T &= \left[\frac{1}{2}(A - A^T) \right]^T \\ &= \frac{1}{2}[(A - A^T)]^T \\ &= \frac{1}{2}(-A + A^T) \\ &= \frac{1}{2}(A^T - A) \\ &= -\frac{1}{2}(A - A^T) \\ &= -A_1 \end{aligned}$$

To prove that $A = A_0 + A_1$ and that $x^T Ax = x^T A_0 x$, it suffices to combine from the previous two parts to yield the following:

$$\begin{aligned} A_0 + A_1 &= \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T) = A \\ \implies x^T Ax &= x^T A_0 x + x^T A_1 x \\ \implies x^T Ax &= x^T A_0 x \end{aligned}$$

The last line above resulted since $x^T A_1 x = 0$ by Problem #17, as desired. $\square$

4.2: LU and Cholesky Factorizations

- [2pts] 5) Prove that an upper or lower triangular matrix is nonsingular if and only if its diagonal elements are all different from 0.

Solution:

Without loss of generality, let U be an upper triangular $n \times n$ nonsingular matrix. Then $UU^{-1} = I$. The following suffices to both directions:

$$\begin{aligned} U \text{ nonsingular} &\iff \text{Det}(U) \neq 0 \\ &\iff u_{11}u_{22} \cdots u_{nn} \neq 0 \\ &\iff \text{each of } a_{11}, a_{22}, \dots, \text{ is nonzero} \end{aligned}$$

Thus, an upper or lower triangular matrix is nonsingular if and only if its diagonal elements are all different from 0. A similar argument can be made for a lower triangular matrix.

□